

Mars-JEPA: Multispectral Joint Embedding Predictive Architectures for Martian Landslide Segmentation

BMVC 2026 Submission # 1133

Abstract

Autonomous Martian exploration relies heavily on geomorphological mapping to mitigate mechanical hazards to robotic assets. While multispectral datasets have improved detection accuracy, traditional supervised models lack generalizability, and generative self-supervised approaches are computationally prohibitive and prone to learning trivial features through pixel reconstruction. In this work, we introduce Spatio-Spectral JEPA (SS-JEPA), a Joint Embedding Predictive Architecture specifically designed for multispectral terrain feature representation learning. Specifically, we propose a novel Spatio-Spectral masking strategy that tokenizes channels independently, forcing the model to learn deep physical correlations between heterogenous spectral bands, in an abstract latent space. On the MMLSv2 dataset, our best-performing SS-JEPA configuration achieves 0.865 mIoU, outperforming I-JEPA baselines by 2.8% and significantly exceeding the performance of single-model supervised backbones. While our single-model backbone trails the current state-of-the-art 5-fold ensemble by a marginal 0.2% mIoU, it does so with a $5.6\times$ reduction in parameter count. Our results demonstrate that latent-space prediction is more robust to Martian sensor noise and varying lighting conditions than traditional generative SSL. More broadly, our work presents a scalable and efficient framework for learning semantic representations from multispectral remote sensing data for terrain feature analysis.

1 Introduction

Autonomous planetary exploration and mapping, specifically on Mars, has generated vast amounts of remote sensing data, massively accelerating our understanding of extra-terrestrial environments. Analysis of the Martian terrain highlights the frequent occurrence of landslides, characterized the mass movement of debris across large areas. This necessitates autonomous geomorphological feature mapping and landslide detection, which is critical for identifying geological shifts and safeguarding robotic and human assets deployed on the surface of the planet. Recent datasets like MMLSv2 [1] have shifted the field from monochromatic analysis to multispectral imagery, stacking bands like visible (RGB), digital elevation model (DEM), slope and thermal inertia together.

Traditional fully supervised learning methods are designed with inductive biases to effectively capture spatial relationships, but struggle to learn spectral correlations in multispectral

imagery. This results in the learned representations failing to capture the multimodal dependencies present in such data, leading to heavy reliance on larger model sizes to overcome the performance barrier [13] on landslide detection tasks. Meanwhile, self-supervised methods [9] can be used to learn rich semantic features, but the usage of generative approaches such as pixel reconstruction from masked inputs [10] limits their computational efficiency for high-dimensional multispectral data. The introduction of the joint embedding predictive architecture (JEPA) aims to solve this by modeling representation learning as a predictive task in the latent space, overcoming the computational complexity associated with pixel-level reconstruction. However, specific adaptations of this framework addressing the challenge of learning semantic representations from multispectral data remain underexplored.

Our primary insight is that for a given spatial region, different spectral bands carry independent but correlated information. To learn rich semantic representations for multispectral data, the model must develop an understanding of these inter-modal relationships. To this end, we introduce Spatio-Spectral JEPA (SS-JEPA), an adaptation of the JEPA framework to learn rich semantic representations from multispectral imagery. Implementing tokenization and masking across both, spatial regions and spectra, the model learns spatial as well as spectral relationships in the data through a latent prediction objective. Our SS-JEPA encoder, paired with a UPerNet [14] segmentation decoder head, achieves performance on par with state-of-the-art fully supervised ensemble models $5.6\times$ larger (20M vs 113M), and beats baseline models with JEPA backbones trained with spatial-only masking, specifically I-JEPA [10], by 2.8% mIoU. Further, qualitative latent visualizations reveal the representations learned by our SS-JEPA to be much richer in semantic information than those learned by the I-JEPA baseline encoder.

Our specific contributions include:

- i) We propose SS-JEPA, a joint-embedding predictive architecture designed for multispectral image encoding via a novel spatial-spectral masking strategy
- ii) We demonstrate that forcing the model to learn cross-spectral latent correlations yields representations that capture rich geological and physical semantics, validated qualitatively through latent visualizations
- iii) We achieve state-of-the-art single-model performance on the MMLSv2 dataset, significantly outperforming supervised baselines while demonstrating extreme parameter and compute efficiency.

2 Related Work

The development of a robust landslide detection system for the Martian surface requires an interdisciplinary understanding of computer vision and planetary science. In this section, we first discuss the geomorphological challenges and history of Martian surface exploration in Section 2.1. We then review the evolution of semantic segmentation frameworks in Section 2.2, followed by a discussion of the paradigm shift from generative models toward predictive self-supervised learning in Section 2.3.

2.1 Planetary Surface Analysis and Exploration

Mars has long been a focal point for planetary exploration, with significant research dedicated to identifying landslides on its surface. Detecting landslides automatically is a non-trivial task due to shifting atmospheric conditions and complex geomorphology. Historically,

experts have identified these features manually using high-resolution imagery from the High Resolution Imaging Science Experiment (HiRISE) [19].

Early automated systems primarily utilized physics-based features derived from digital elevation models, such as slope and elevation. Recently, the field has transitioned towards deep learning, employing Convolutional Neural Networks (CNNs) [20] for automated mapping. While traditional studies relied on monochromatic imagery, the introduction of the MMLSv2 dataset [21] enabled the use of multimodal data, combining spectral imagery with topographic information to improve detection accuracy.

2.2 Advancements in Semantic Segmentation

Semantic Segmentation has advanced from basic pixel-wise classifiers to encoder-decoder frameworks designed to capture multi-scale context. Early designs like U-Net [22] became highly popular because of its simple design based on skip connections that preserved spatial features. While U-Net++ [23] improved this by using nested and dense skip pathways to bridge the semantic gap between the features. To handle objects at different scales, Feature Pyramid Network (FPN) [18] and Pyramidal Attention Networks (PAN) [16] focussed on integrating high-level global semantics with low-level spatial features.

Further refinements led to the development of modules like DeepLabv3+ [6] which relied on atrous convolutions to expand receptive fields. More recently attention driven architectures like MANet [17] and UPerNet [23] have sought to capture global dependencies effectively. Most recently transformer based frameworks like SegFormer [24] showed the efficiency of hierarchical self-attention in capturing long range concept.

2.3 Self-Supervised Learning in Remote Sensing

Self-supervised representation learning [6] has seen a paradigm shift from contrastive frameworks, which rely heavily on data augmentations, to Masked Image Modeling (MIM) [25]. While generative approaches like [2, 10] have shown promising results by reconstructing raw pixels, their scalability is limited by the high computational complexity of such pixel-level reconstruction tasks, besides introducing additional computational overhead trying to model low-level noise in the data. The Joint Embedding Predictive Architecture (JEPA) [15] and subsequent image encoding frameworks like I-JEPA [11] shift the paradigm to learning representations by predicting latents of masked regions instead of pixel-level reconstruction. This overcomes the high computational overhead and sensitivity to low-level noise faced in reconstruction-based SSL. Although JEPA encoders have been used for remote sensing for sparse retrieval tasks [2, 12] or on visible bands only [14], its application to multispectral image encoding for dense prediction tasks like landslide segmentation remains unexplored.

3 Methodology

In this work, we study the downstream performance on segmentation tasks for two JEPAs trained with different tokenization masking strategies, on multispectral images.

3.1 Problem Formulation

Consider a multispectral image $I \in \mathbb{R}^{C \times H \times W}$, where C is the fixed number of bands, H, W are the height and width of the image respectively. We define a binary segmentation mask $M \in \{0, 1\}^{H \times W}$ over the pixels in the image, where a pixel $M[i, j]$ at row i and column j in the segmentation mask M has a value of 1 if it is predicted to be affected by a landslide, and 0 otherwise.

The Martian landslide detection problem is to develop a segmentation model $S_\gamma: \mathbb{R}^{C \times H \times W} \rightarrow \{0, 1\}^{H \times W}$, parameterized by γ , to satisfy the objective function

$$\max_{\gamma} \mathbb{E}_{(I, M) \sim D} [\sigma_{\text{sim}}(S_\gamma(I), M)], \quad (1)$$

where the multispectral image and corresponding segmentation mask pairs (I, M) are sampled from the dataset D , and $\sigma_{\text{sim}}(M_1, M_2) \in \mathbb{R}$ is a similarity metric between two binary segmentation masks M_1 and M_2 . The aim is to maximize the expected similarity between the predicted segmentation mask and the actual segmentation mask, for all examples in the dataset, by performing an optimization over the model parameters γ .

3.2 Joint Embedding Predictive Architecture (JEPA)

The Joint Embedding Predictive Architecture (JEPA) trains an encoder by applying inductive biases to maximize the information content about the input. In the context of multispectral image encoding, it does so by training an encoder to extract such features from a perturbed version of the image, that enable a lighter predictor module to predict the features of the unperturbed version of the image, conditioned on the perturbation applied to image to form the context. The perturbed image is called the context x , the unperturbed image is called the target y , and the respective features extracted by the encoder E_θ are denoted by s_x and s_y . The predictor module, denoted by D_ϕ , conditioned on the perturbation z , outputs a prediction for the target features, denoted by $\hat{s}_y$. Mathematically,

$$s_x = E_\theta(x) \quad (2)$$

$$s_y = E_\theta(y) \quad (3)$$

$$\hat{s}_y = D_\phi(s_x, z) \quad (4)$$

$$\mathcal{L}_{\text{pred}} = \mathcal{D}(\hat{s}_y, s_y) \quad (5)$$

$$\theta^*, \phi^* = \arg \min_{\theta, \phi} \mathcal{L}_{\text{pred}}. \quad (6)$$

In this formulation, Eqs. (2) and (3) define the latent encoding of the context and target signals. Equation (4) describes the predictive step in the latent space. The discrepancy between the predicted and actual target embeddings is quantified in Eq. (5) by a distance metric $\mathcal{D}(\cdot, \cdot)$. Finally, Eq. (6) defines the optimization objective, where the parameters θ and ϕ are learned by minimizing the prediction error in the latent space. Finally, to prevent latent space collapse in the encoder, we apply the SIGReg regularization [1] loss to both, the context and target features, s_x and s_y , calculating the total loss as

$$\mathcal{L}_{\text{total}} = (1 - \lambda) \mathcal{L}_{\text{pred}} + \lambda (\mathcal{L}_{\text{SIGReg}}(s_x) + \mathcal{L}_{\text{SIGReg}}(s_y)), \quad (7)$$

where $\lambda \in [0, 1]$ is a hyperparameter. In Eq. (5), we use the L_2 divergence as the predictive loss divergence metric $\mathcal{D}$.

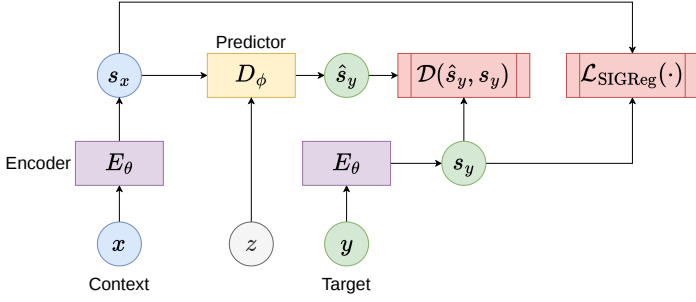


Figure 1: Overview of the JEPA architecture

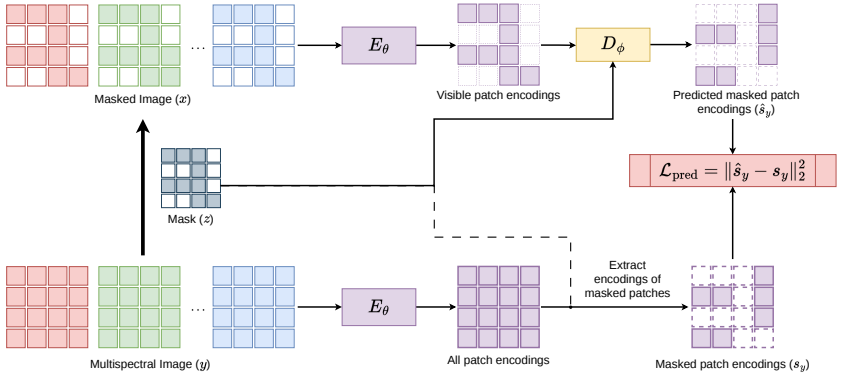


Figure 2: Masking strategy and predictive loss calculation for the I-JEPA encoder

3.3 I-JEPA Baseline for Multispectral Image Encoding

We benchmark the I-JEPA [14] framework for learning semantic representations for multispectral images, treating multispectral images like regular RGB images with extra channels. As shown in Fig. 2, the context x is constructed by applying the same patch-wise mask to all channels, while the target y is the unmasked image, containing information from all patches and channels. We implement the encoder E_θ as a vision transformer (ViT) [8] with learnable patch positional embeddings. The predictor is implemented as a transformer decoder with the positional embeddings of the masked patches as the query and the encodings for the visible patches as the key and value. The output is the prediction for encodings of the masked patches. The predictive loss is calculated by matching the predicted and actual encodings for the masked patches.

3.4 Spatio-Spectral JEPA (SS-JEPA)

To learn stronger cross-modal dependencies in the semantic representations, we introduce Spatio-Spectral JEPA (SS-JEPA), which implements channel-wise masking and tokenization in contrast to the common mask and tokenization strategy across all channels, from Section 3.3. As shown in Fig. 3, SS-JEPA uses distinct masks per channel, and the en-

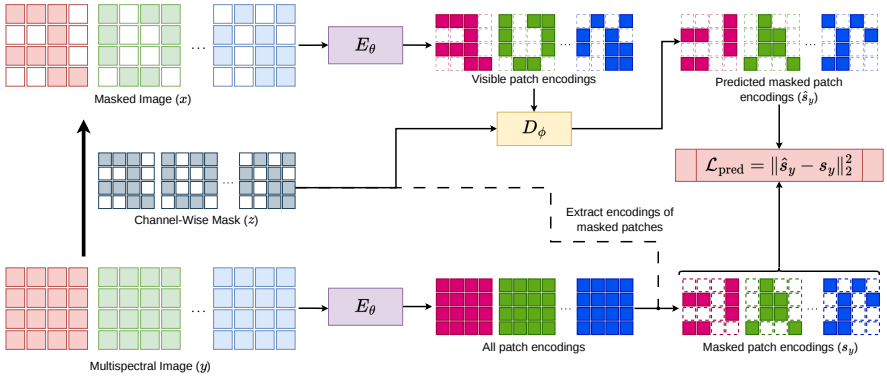


Figure 3: Masking strategy and predictive loss calculation for the SS-JEPA encoder

coder outputs patch encodings for each visible patch in each channel. The encoder E_θ is implemented as a modified ViT that tokenizes each patch in each channel, unlike the common tokenizer for all channels in I-JEPA. The predictor is a transformer decoder, with the spatial-spectral positional embeddings of the masked patches as queries and the encodings for the visible patches as input. The output is the prediction for the encodings of the masked patches. The predictive loss is calculated as the L_2 divergence between the predicted and actual encodings for the masked patches.

Note, here, that a “patch” above is associated with a particular channel, unlike I-JEPA. Additionally, the spatial-spectral positional embeddings for each patch is obtained by adding the patch positional embedding for that patch with the corresponding *channel* embedding for that patch. Mathematically,

$$p_{c,j} = \mathbf{P}_{\text{spatial}}[j] + \mathbf{P}_{\text{spectral}}[c], \quad (8)$$

where $p_{c,j}$ is the spatial-spectral positional embedding for channel c , for the spatial location indexed by j . Here, $\mathbf{P}_{\text{spatial}} \in \mathbb{R}^{n_p \times d}$ and $\mathbf{P}_{\text{spectral}} \in \mathbb{R}^{n_c \times d}$ are the spatial and spectral embedding matrices respectively, and n_p and n_c are the number of patches and channels in each image, respectively. In our dataset, there are $n_c = 7$ bands in the multispectral images.

3.5 Segmentation Models with Self-Supervised Backbones

Following the self-supervised pre-training phase, we integrate the JEPA encoders with a supervised segmentation decoder head to construct a segmentation model. As illustrated in Fig. 4, the primary distinction between the two architectures lies in the nature of the latent features provided to the segmentation head S_{seg} . While I-JEPA generates spectrally aggregated encodings, SS-JEPA preserves the modality-specific information through a channel-wise feature representation. To facilitate this integration, we construct a multiscale feature

pyramid with features

$$f_0 = I \quad (9)$$

$$f_1 = \mathcal{P}_{2 \times 2}^{\max}(f_0) \quad (10)$$

$$f_2 = \mathcal{P}_{2 \times 2}^{\max}(f_1) \quad (11)$$

$$f_3 = E_{\theta}(I) \quad (12)$$

$$f_4 = \mathcal{P}_{2 \times 2}^{\max}(f_3) \quad (13)$$

$$f_5 = \mathcal{P}_{2 \times 2}^{\max}(f_4), \quad (14)$$

where $\mathcal{P}_{2 \times 2}^{\max}$ denotes a max-pooling operation with a 2×2 kernel, I is an image sampled from the dataset, and E_{θ} is the encoder being integrated. In this hierarchy, f_0 , f_1 , and f_2 serve as high resolution skip connections that preserve sharp topographic edges. The bottleneck layer, f_3 is the point of divergence between the two pre-training strategies. In the I-JEPA path, the encoder treats the multispectral input as a unified spatial cube, resulting in a single feature map where spectral information is collapsed into the channel dimension whereas the SS-JEPA path maintains the integrity of the individual modalities. It outputs a set of n_c independent feature maps of shape $d \times h \times w$ representing the encodings for each modality as a feature map, where for an image I of resolution $H \times W$ pixels tokenized into patches of size $p \times p$, $h = H/p$ and $w = W/p$. These maps are subsequently processed by a spectral fusion module W_{fuse} which fuses the encoding for a particular patch across all channels, resulting in a d -dimensional feature map of shape $h \times w$. This spatio-spectral input ensures that the segmentation head S_{seg} receives context-rich features that account for the cross-modal relations learned during pre-training. Finally, f_4 and f_5 provides the global context required for consistent landslide identification across diverse martian surface.

The segmentation model is trained using a hybrid objective function that combines region-level overlap consistency with pixel-wise classification accuracy. To maximize the similarity between the predicted segmentation mask $\hat{M}$ and the ground-truth mask M , the overall segmentation loss is defined as

$$\mathcal{L}_{seg} = \mathcal{L}_{Dice} + \mathcal{L}_{BCE} \quad (15)$$

$$\mathcal{L}_{Dice} = 1 - \frac{2 \sum_{i=1}^N \hat{M}_i M_i + \varepsilon}{\sum_{i=1}^N \hat{M}_i + \sum_{i=1}^N M_i + \varepsilon} \quad (16)$$

$$\mathcal{L}_{BCE} = -\frac{1}{N} \sum_{i=1}^N [M_i \log(\hat{M}_i) + (1 - M_i) \log(1 - \hat{M}_i)], \quad (17)$$

where $\hat{M}_i \in [0, 1]$ denotes the predicted probability for the i^{th} pixel belonging to the foreground class, while $M_i \in \{0, 1\}$ represents the corresponding ground-truth binary label. The variable $N = HW$ denotes the total number of pixels in the segmentation mask. The Dice loss, denoted by $\mathcal{L}_{Dice}$, measures the structural overlap between the predicted and ground-truth masks, encouraging global region consistency. The binary cross-entropy loss, denoted by $\mathcal{L}_{BCE}$ evaluates pixel-wise classification accuracy by penalizing incorrect foreground and background predictions. The smoothing constant ε is introduced to avoid numerical instability and division-by-zero during optimization.

During the downstream segmentation task, the parameters of the encoder E_{θ} are kept frozen in order to evaluate the quality and transferability of the learned self-supervised rep-

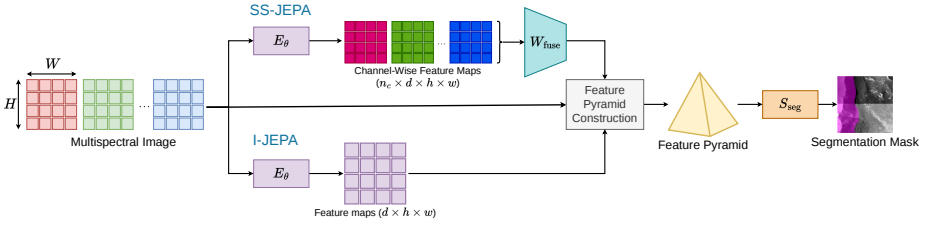


Figure 4: Integration of JEPA encoder backbones with segmentation heads

representations. Only the parameters of the segmentation decoder are optimized using the supervised segmentation objective defined above. The supervised fine-tuning process is performed on the training subset of the MMLSv2 dataset [24].

This two-stage training strategy, consisting of self-supervised pre-training followed by supervised segmentation, enables the model to effectively exploit the large volume of unlabeled Martian surface imagery. As a result, the learned representations improve the robustness and generalization capability of the final landslide detection framework, particularly in scenarios with limited annotated data.

4 Experimental Evaluation and Results

In this section, we provide a comprehensive empirical evaluation of the Mars-JEPA framework landslide segmentation from multispectral images. We experimentally evaluate the quality of the semantic representations from encoders and validate the superiority of representations from our SS-JEPA, adapted to the multispectral image domain.

4.1 Experimental Setup

The proposed framework described in Section 3 was evaluated on the MMLSv2 dataset [24], a multimodal Martian landslide segmentation benchmark consisting of co-registered 128×128 GeoTIFF tiles. Each sample contains seven heterogeneous sensing channels, including Thermal Inertia (THEMIS, ~ 100 m/px), Slope maps derived from DEM data, Digital Elevation Models (MOLA/HRSC, ~ 200 m/px), Grayscale imagery from CTX (~ 6 m/px), and a three-channel RGB basemap generated from Viking mosaics (~ 232 m/px).

The publicly available split of the dataset contains 465 labeled training samples and 66 labeled validation samples. Owing to the heterogeneous nature of the sensing modalities and the varying numerical distributions across channels, all multispectral inputs were pre-processed using channel-wise normalization prior to training. All training and inference was performed on a single NVIDIA RTX A4500 GPU with 20GB VRAM.

4.2 Implementation Details

The proposed SS-JEPA framework was trained for 150 epochs using the AdamW optimizer with a cosine annealing with linear warmup learning rate scheduler. To improve the robustness and generalization capability of the learned spatio-spectral representations across heterogeneous multispectral modalities, several stochastic spatial augmentations were applied

during training. These augmentations included random horizontal and vertical flipping, random rotation in the range $[-90^\circ, 90^\circ]$, random resized cropping, and additive Gaussian noise $x_{\text{noise}} \sim \mathcal{N}(0, 0.1)$.

During self-supervised pre-training, we apply linear scheduling on the masking ratio in the range $[0.5, 0.85]$. Further, the SIGReg [9] regularization hyperparameter λ from Eq. (7) was set to 0.3. The above hyperparameter values were empirically observed to be optimal.

Following self-supervised pre-training, the learned encoder weights were transferred to downstream segmentation architectures for semantic landslide segmentation. The segmentation heads were trained for 70 epochs with a learning rate of 3×10^{-4} with the same optimizer and learning rate scheduler as the pre-training phase.

Unlike the self-supervised pre-training stage, no image augmentations were applied during segmentation-head training. The segmentation performance was quantitatively evaluated using precision, recall, F1-score, foreground IoU, background IoU, and mean Intersection-over-Union (mIoU).

4.3 Backbone and Decoder Compatibility

The comparative analysis of segmentation performance across different encoder backbones is presented in Table 1. Our results demonstrate that the proposed SS-JEPA architecture, when paired with a UPerNet [23] decoder, achieves the highest performance with an mIoU of 0.865. This represents a 2.8% improvement over the best I-JEPA configuration, which utilizes a SegFormer [24] head to reach 0.837 mIoU. The consistent performance gain observed with the SS-JEPA backbone validates the superior effectiveness of the spatio-spectral masking strategy over the standard spatial-only masking approach.

The observed performance gap is primarily attributed to the differences in the predictive constraints imposed during self-supervised pre-training. In the multispectral I-JEPA implementation, the application of an identical spatial mask across all n_c channels restricts the encoder to learning representations based on spatial proximity. In this configuration, if a region is masked, the model lacks information across all spectral modalities for that location, forcing the predictor to rely solely on surrounding spatial context. In contrast, the channel-wise masking in SS-JEPA introduces an inter-modal predictive task. Because patches are masked independently per channel, the encoder must learn to reconstruct latent features in one spectral band (*e.g.*, thermal inertia) using visible context from other bands (*e.g.*, digital elevation model or visible bands). This forces the model to internalize the underlying physical and geological correlations between different Martian surface modalities, resulting in more descriptive semantic embeddings, resulting in improved segmentation performance.

4.4 Comparison with State-of-the-Art

Our SS-JEPA + UPerNet model achieves 0.865 mIoU, defining the Pareto frontier in Fig. 5. It outperforms standard supervised baselines like U-Net++ [26], PSPNet [25], DeepLabV3+ [8], and SegFormer [24] which plateau between 0.81 and 0.83, proving fully supervised training from scratch cannot fully exploit complex cross-modal correlations. At just 20M parameters, our single-model framework is $5.6\times$ smaller than the 113M parameter DualSwinV2 5-fold ensemble [13], providing highly competitive performance, trailing the latter by just 0.2% mIoU. It also beats the previous best single-model baseline, DualSwinV2-S [13], by 3.7% mIoU and the 33M parameter TinySwin + FPN by 2.8% mIoU. Transitioning from I-JEPA to SS-JEPA yields a 15.6% parameter reduction alongside a 2.8% mIoU gain, confirming

Encoder	Segmentation	MIoU	Precision	Recall	F1 Score	FG IOU	BG IOU
I-JEPA	UPerNet [13]	0.825	0.861	0.868	0.864	0.774	0.876
	UNet [14]	0.829	0.853	0.872	0.862	0.780	0.878
	UNet++ [15]	0.825	0.860	0.880	0.869	0.777	0.873
	FPN [16]	0.825	0.849	0.890	0.869	0.779	0.871
	MANet [17]	0.810	0.845	0.868	0.856	0.761	0.859
	DeepLabv3+ [8]	0.818	0.858	0.869	0.863	0.766	0.871
	PAN [18]	0.814	0.856	0.869	0.862	0.761	0.867
	SegFormer [19]	0.837	0.858	0.876	0.865	0.790	0.884
SS-JEPA	UPerNet [13]	0.865	0.895	0.907	0.901	0.820	0.910
	UNet [14]	0.852	0.863	0.887	0.875	0.824	0.880
	UNet++ [15]	0.840	0.877	0.860	0.868	0.807	0.873
	FPN [16]	0.827	0.866	0.878	0.872	0.776	0.878
	MANet [17]	0.826	0.869	0.870	0.870	0.782	0.870
	DeepLabv3+ [8]	0.820	0.858	0.866	0.862	0.769	0.871
	PAN [18]	0.817	0.862	0.859	0.860	0.764	0.870
	SegFormer [19]	0.834	0.800	0.848	0.814	0.786	0.882

Table 1: Segmentation performance with self-supervised encoder backbones

Model	MIoU	Precision	Recall	F1 Score	FG IOU	BG IOU
U-Net [14]	0.814	0.858	0.868	0.863	0.759	0.868
U-Net++ [15]	0.823	0.864	0.879	0.871	0.772	0.875
PSPNet [16]	0.828	0.866	0.884	0.875	0.778	0.878
DeepLabV3 [8]	0.817	0.870	0.860	0.865	0.763	0.872
DeepLabV3+ [8]	0.829	0.863	0.889	0.876	0.779	0.878
SegFormer [19]	0.812	0.859	0.863	0.861	0.756	0.867
DualSwinV2 + U-Net++, 5-fold ensemble [13]	0.867	0.900	0.911	0.905	0.827	0.907
DualSwinV2-S + Hybrid Decoder [19]	0.828	0.856	0.899	0.877	-	-
SS-JEPA + UPerNet (Ours)	0.865	0.895	0.907	0.901	0.820	0.910
I-JEPA + SegFormer (Ours)	0.837	0.858	0.876	0.865	0.790	0.884

Table 2: Performance comparison between different methodologies

that our spatio-spectral masking strategy internalizes inter-modal structures more effectively than spatial-only SSL.

Class-specific metrics in Table 2 validate this architectural reliability. SS-JEPA secures the highest background IoU (0.910), reflecting superior noise rejection, alongside a foreground IoU of 0.820 with balanced precision-recall. These results demonstrate that our SS-JEPA captures richer geological semantics than standard supervised alternatives. By minimizing inference overhead while preserving robust representations, this single-pass framework replaces heavy, computationally expensive ensembles, posing as a suitable candidate for the low-power hardware typical of planetary exploration platforms and Martian orbiters.

4.5 Channel Ablation Study

To isolate and study the contribution of individual channels, we evaluate SS-JEPA pre-trained on different subsets of channels, with a UPerNet [23] segmentation decoder, in Table 3.

Utilizing purely optical channels (RGB + Grayscale) establishes a baseline performance of 0.812 mIoU. This metric highlights a persistent limitation in planetary surface imaging, where atmospheric dust variations and uniform albedo profiles frequently obscure landform boundaries. In contrast, isolating structural geometry channels (DEM + Slope) improves performance to 0.835 mIoU. This margin demonstrates that topographic markers, specifically the abrupt elevation drops found at source scarps and the adjacent depositional accumulation zones, provide highly discriminative features that remain entirely invariant to surface

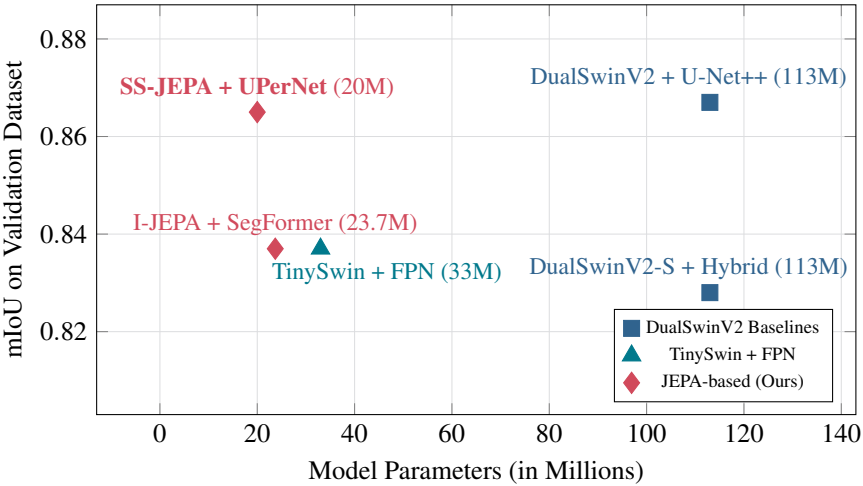


Figure 5: Performance vs. model size comparison

RGB	Gray	DEM	Slope	Thermal	mIoU
✓	✓				0.812
		✓	✓		0.835
✓	✓	✓			0.841
✓	✓	✓	✓		0.854
		✓	✓	✓	0.846
✓	✓	✓	✓	✓	0.865

Table 3: Channel Ablation Study of the SS-JEPA + UPerNet Segmenation Model Configuration

illumination or optical noise. Progressive cross-modal integration yields consistent performance gains, with the full 7-channel configuration achieving a peak score of 0.865 mIoU. While the thermal inertia modality exhibits a lower native spatial resolution, its integration with geometric features provides a 0.011 mIoU performance boost. These results demonstrate that combining optical, topographic, and thermophysical data allows our framework to extract complementary physical structures, maximizing landslide segmentation accuracy across heterogeneous Martian terrains.

4.6 Qualitative Results and Representation Analysis

This section provides a qualitative analysis of the segmentation performance of models using the JEPA encoder backbones, and an evaluation of the latent representations learned by the JEPA encoders.

Segmentation Performance A qualitative comparison of segmentation masks predicted by the best-performing models from Section 4.3 is presented in Fig. 7. The first four columns

Model	Silhouette $\uparrow$	Davies-Bouldin $\downarrow$	Calinski-Harabasz $\uparrow$
I-JEPA	0.017	5.251	521.642
SS-JEPA (Ours)	0.090	2.699	3778.412

Table 4: Evaluation of Latent Encodings using Clustering-based Metrics

present the different multispectral input bands (RGB, digital elevation model (DEM), slope and thermal inertial map), and the last three columns compare the segmentation masks from I-JEPA + SegFormer, SS-JEPA + UPerNet and the ground truth from the dataset respectively, overlayed on the grayscale band. The SS-JEPA + UPerNet model shows less false positives and produces segmentation masks with boundaries tighter to the ground truth, whereas those of I-JEPA are spread outward. We attribute this to the sharper and richer semantic representations learned by the SS-JEPA encoder, as discussed further.

Latent Representation Analysis A visualization of the latent encodings for patches of different multispectral images from the I-JEPA and SS-JEPA encoders is presented in Fig. 8, where the first three principal components of the patch embeddings are normalized and rendered as pseudo-RGB images. Compared to I-JEPA, the proposed SS-JEPA encoder produces spatially coherent and semantically structured latent representations, preserving larger terrain-level patterns and cross-spectral relationships across neighboring patches. In contrast, I-JEPA embeddings exhibit fragmented, high-frequency activations with weaker spatial consistency.

To further quantify representation quality, we evaluate the latent embeddings using clustering-based metrics, reported in Table 4. SS-JEPA achieves a substantially higher Silhouette score (0.090 vs. 0.017) and Calinski-Harabasz score (3778.412 vs. 521.642), while also reducing the Davies-Bouldin index (2.699 vs. 5.251). These results indicate that SS-JEPA produces significantly more compact and better-separated latent clusters, suggesting improved semantic organization within the embedding space.

Additionally, the singular value spectrum shown in Fig. 6 demonstrates that SS-JEPA maintains a broader eigenspectrum with slower singular value decay compared to I-JEPA. This indicates improved utilization of latent dimensions and richer feature diversity, reducing representation collapse while preserving meaningful multispectral structure. We attribute these improvements to the proposed spatial-spectral masking strategy, which encourages the encoder to jointly model spatial continuity and spectral relationships across channels. Such semantically organized latent representations are critical for achieving strong downstream landslide segmentation performance under constrained parameter budgets, as demonstrated earlier in Section 4.4.

5 Conclusion

In this work, we introduce a Spatio-Spectral Joint Embedding Predictive Architecture (SS-JEPA) specifically designed for the challenges of multimodal Martian landslide segmentation. By shifting from traditional end-to-end supervised learning methods to a latent-predictive representation learning framework, our approach effectively models the physical correlations between spectral bands.

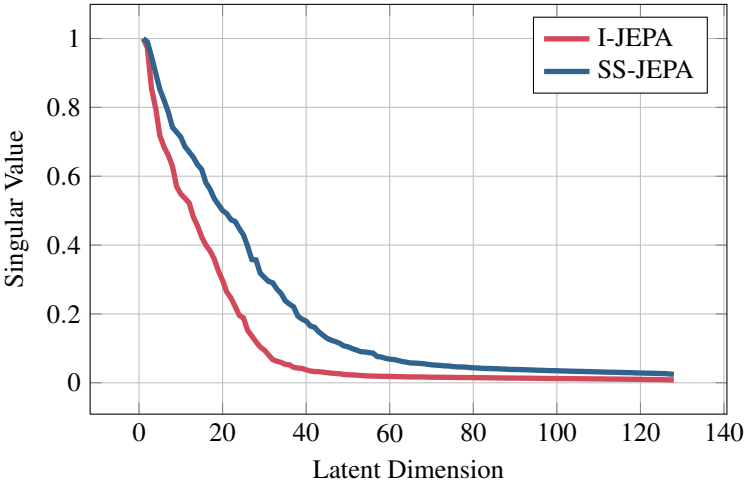


Figure 6: Singular Values of Encodings from I-JEPA vs. SS-JEPA

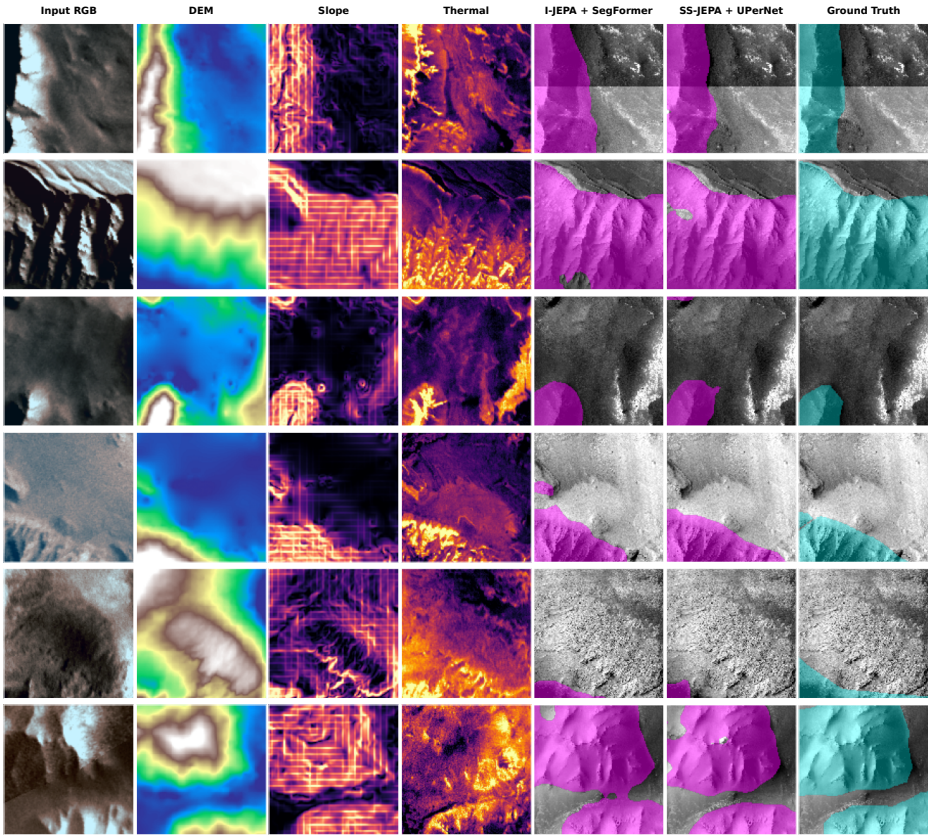


Figure 7: Comparison of masks from segmentation models with JEPA encoder backbones

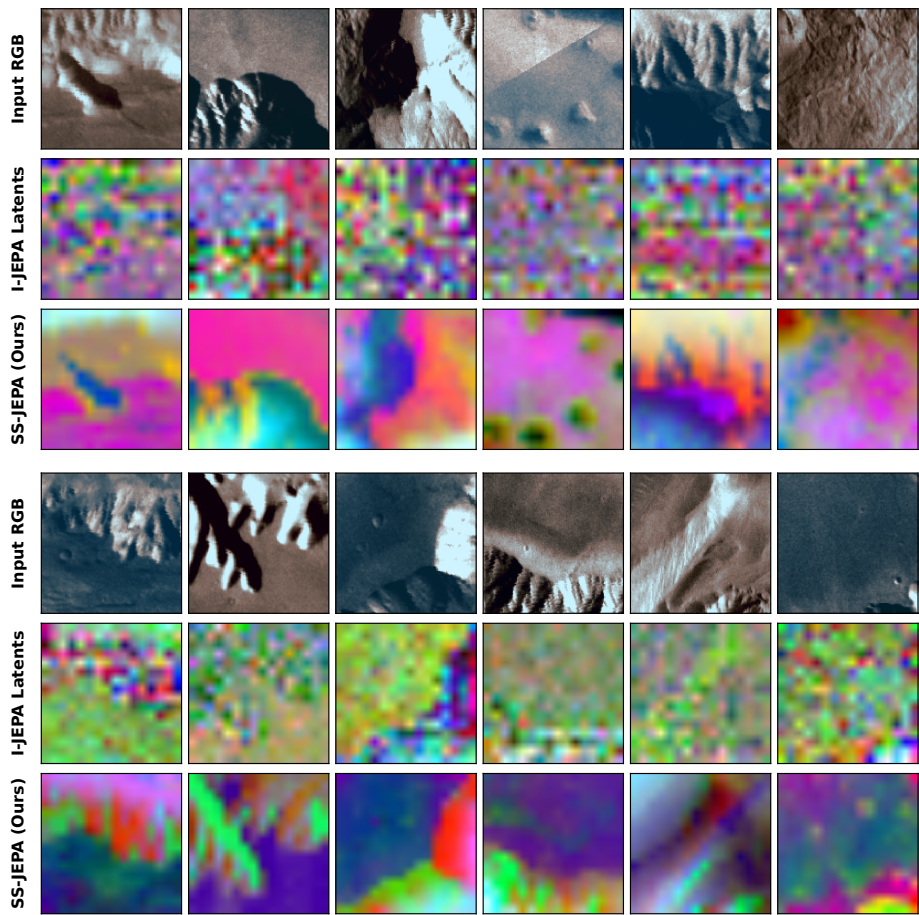


Figure 8: Visualization of patch latent encodings from JEPA encoders

Our primary contribution is a spatio-spectral masking strategy that enables the model to learn deep physical correlations between heterogeneous modalities. We achieve landslide segmentation performance on par with SOTA methods that use $5.6\times$ larger models than ours. Beyond these numerical gains, the proposed architecture captures the intrinsic physical correlations between different bands in multispectral terrestrial imagery to understand how the interaction between them shapes the Martian landscape, through qualitative visual analysis. Despite these results, the scarcity of labeled Martian data remains a constraint, necessitating further validation across broader planetary datasets to ensure global generalizability.

The learned representations further highlight the importance of cross-modal interactions in multispectral planetary imagery, demonstrating how geological structures can manifest differently across spectral bands while still preserving meaningful latent relationships. Overall, this work presents a robust, scalable, and computationally efficient approach for automated Martian landslide segmentation, providing a practical pathway toward intelligent planetary exploration systems and protecting crucial assets deployed on the surface of the Red Planet.

References

[1] Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 15619–15629, 2023.

[2] Guillaume Astruc, Nicolas Gonthier, Clement Mallet, and Loic Landrieu. Omnisat: Self-supervised modality fusion for earth observation. In *European Conference on Computer Vision*, pages 409–427. Springer, 2024.

[3] Randall Balestriero and Yann LeCun. Lejepa: Provable and scalable self-supervised learning without the heuristics, 2025. URL <https://arxiv.org/abs/2511.08544>.

[4] Liang-Chieh Chen, George Papandreou, Iasonas Kokkinos, Kevin Murphy, and Alan L. Yuille. Deeplab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected crfs, 2017. URL <https://arxiv.org/abs/1606.00915>.

[5] Liang-Chieh Chen, Yukun Zhu, George Papandreou, Florian Schroff, and Hartwig Adam. Encoder-decoder with atrous separable convolution for semantic image segmentation. *CoRR*, abs/1802.02611, 2018. URL <http://arxiv.org/abs/1802.02611>.

[6] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey E. Hinton. A simple framework for contrastive learning of visual representations. *CoRR*, abs/2002.05709, 2020. URL <https://arxiv.org/abs/2002.05709>.

[7] Shabnam Choudhury, Yash Salunkhe, Sarthak Mehrotra, and Biplab Banerjee. Rejepa: A novel joint-embedding predictive architecture for efficient remote sensing image retrieval. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 2373–2382, 2025.

[8] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021. URL <https://openreview.net/forum?id=YicbFdNTTy>.

[9] Linus Ericsson, Henry Gouk, Chen Change Loy, and Timothy M. Hospedales. Self-supervised representation learning: Introduction, advances, and challenges. *IEEE Signal Processing Magazine*, 39(3):42–62, May 2022. ISSN 1558-0792. doi: 10.1109/msp.2021.3134634. URL <http://dx.doi.org/10.1109/MSP.2021.3134634>.

[10] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 16000–16009, 2022.

- [11] Vlad Hondru, Florinel Alin Croitoru, Shervin Minaee, Radu Tudor Ionescu, and Nicu Sebe. Masked image modeling: A survey. *International Journal of Computer Vision*, 133(10):7154–7200, 2025. ISSN 1573-1405. doi: 10.1007/s11263-025-02524-1. URL <http://dx.doi.org/10.1007/s11263-025-02524-1>.
- [12] Aniket Junghare and Biplab Banerjee. Ua-jepa: Uncertainty-aware joint-embedding predictive learning for remote sensing image retrieval. 2026. doi: 10.2139/ssrn.6592855. URL <http://dx.doi.org/10.2139/ssrn.6592855>.
- [13] Shahriar Kabir, Abdullah Muhammed Amimul Ehsan, Istiak Ahmmmed Rifti, and Md Kaykobad Reza. Dualswinfusionseg: Multimodal martian landslide segmentation via dual swin transformer with multi-scale fusion and unet++, 2026. URL <https://arxiv.org/abs/2603.14132>.
- [14] Kursat Komurcu and Linas Petkevicius. Sat-jepa-diff: Bridging self-supervised learning and generative diffusion for remote sensing. *arXiv preprint arXiv:2603.13943*, 2026.
- [15] Yann LeCun et al. A path towards autonomous machine intelligence version 0.9. 2, 2022-06-27. *Open Review*, 62(1):1–62, 2022.
- [16] Hanchao Li, Pengfei Xiong, Jie An, and Lingxue Wang. Pyramid attention network for semantic segmentation. *CoRR*, abs/1805.10180, 2018. URL <http://arxiv.org/abs/1805.10180>.
- [17] Rui Li, Shunyi Zheng, Ce Zhang, Chenxi Duan, Jianlin Su, Libo Wang, and Peter M. Atkinson. Multiattention network for semantic segmentation of fine-resolution remote sensing images. *IEEE Transactions on Geoscience and Remote Sensing*, 60: 1–13, 2022. ISSN 1558-0644. doi: 10.1109/tgrs.2021.3093977. URL <http://dx.doi.org/10.1109/TGRS.2021.3093977>.
- [18] Tsung-Yi Lin, Piotr Dollár, Ross B. Girshick, Kaiming He, Bharath Hariharan, and Serge J. Belongie. Feature pyramid networks for object detection. *CoRR*, abs/1612.03144, 2016. URL <http://arxiv.org/abs/1612.03144>.
- [19] Alfred S McEwen et al. Mars reconnaissance orbiter’s high resolution imaging science experiment (hirise). *Journal of Geophysical Research: Planets*, 112(E5), 2007.
- [20] Keiron O’Shea and Ryan Nash. An introduction to convolutional neural networks, 2015. URL <https://arxiv.org/abs/1511.08458>.
- [21] Sidike Paheding, Abel Reyes-Angulo, Leo Thomas Ramos, Angel D. Sappa, A. Rajaneesh, P. B. Hiral, K. S. Sajin Kumar, and Thomas Oommen. Mmlsv2: A multimodal dataset for martian landslide detection in remote sensing imagery. In *Proceedings of IEEE/CVF Computer Vision and Pattern Recognition (CVPR) Workshops*, 2026.
- [22] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In Nassir Navab, Joachim Hornegger, William M. Wells, and Alejandro F. Frangi, editors, *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, pages 234–241, Cham, 2015. Springer International Publishing. ISBN 978-3-319-24574-4.

[23] Tete Xiao, Yingcheng Liu, Bolei Zhou, Yuning Jiang, and Jian Sun. Unified perceptual parsing for scene understanding. *CoRR*, abs/1807.10221, 2018. URL <http://arxiv.org/abs/1807.10221>.

[24] Enze Xie, Wenhai Wang, Zhiding Yu, Anima Anandkumar, Jose M. Alvarez, and Ping Luo. Segformer: Simple and efficient design for semantic segmentation with transformers. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, volume 34, pages 12077–12090. Curran Associates, Inc., 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/file/64f1f27bf1b4ec22924fd0acb550c235-Paper.pdf.

[25] Hengshuang Zhao, Jianping Shi, Xiaojuan Qi, Xiaogang Wang, and Jiaya Jia. Pyramid scene parsing network, 2017. URL <https://arxiv.org/abs/1612.01105>.

[26] Zongwei Zhou, Md Mahfuzur Rahman Siddiquee, Nima Tajbakhsh, and Jianming Liang. Unet++: Redesigning skip connections to exploit multiscale features in image segmentation. *IEEE Transactions on Medical Imaging*, 2019.